Common-pool theories and their corresponding game-theoretical models (see Appendix B) have special importance for the interpretation of PPP games. It is important, however, to see the significant difference between *common pools* supplied by Nature (such as a hunting ground or a meadow) and the goods (such as a road or a fish-pond) *created* and owned by a community.

The social capital of the community—a 'soft' value defined in the theory of institutions—is embodied as 'hardware' in some common goods such as *public road* assets. Financing the public road infrastructure is an issue of creating and/or maintaining *social capital*. Some groups are deeply interested in consumption; others are devoted to accumulation. Fair groups seek a balance between consumption and accumulation. In road infrastructure games it makes sense to identify consumer groups with regard to their readiness to consume or accumulate social capital. The overlap between the sets of criteria to successfully deliver a PPP project, to maintain and increase social capital in a community and to manage a *pro* referendum for an optional tax to finance a public transport infrastructure facility is impressive.

Trends, expectations and areas of further research

PPP is expected to remain an important instrument of transportation policy implementation, political power enforcement and financial capital investment. Its proliferation is in harmony with the continuing deconstruction of public administration and with the popularity of mortgage plans in which individuals, communities and generations leave debts to their descendants. Alternative financing techniques are important because they give governments at any level new options to access available revenues and to smooth and leverage them to meet the expenditures over the time period until public resources become available. Social sciences may clarify the qualitative and quantitative consequences of this trend.

More disciplined public asset accounting (Kadlec and McNeil, 2001) and elucidation of the differences between services and assets can always enhance more skilled public participation in PPP schemes. In general, the better understanding of PPP games can reduce the losses and costs, even if the outcomes are not governed by consensual moral or free-market principles. The legal environment of PPP activity has to be improved. In particular, amendments are needed to regulate the evaluation phase to achieve as much transparency and competition as possible.

The reallocation of roles in the transportation infrastructure development industry is a challenge for education. Engineers have to expand their knowledge (Pethtel and Scharle, 2001). Pure professional technical skills are becoming undervalued and inadequate to influence the outcomes governed by other disciplines' paradigms.

Further analysis of delivered PPP projects (whether successful or failed) is needed to develop an adequate terminology, description techniques and methodology for supporting the delivery of future projects. Case studies of PPP games can be developed for training and teaching purposes and used to enhance the better understanding of partnership issues.

The experience and lessons learned in European PPP practice during the last two decades have to be analysed to identify several particular business-culture-dependent alternatives and trends that are different from those present in the USA.

Specific interdisciplinary research may be focused on the:

- possible conflicts between PPP delivery and standard controlling mechanisms (they

may become more costly and technically more difficult, e.g. contractors may reject even the access of a surveyor);
- evaluation of risks allocated to the public entity (e.g. geo-technical risks, environmental acceptability, subsurface conditions);
- more serious penalties occurring in legal actions with little room for consideration of technical circumstances.

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Notes

1. The wording used in the report is greatly influenced by the approach followed by Shubik in his principal book *Games for Society, Business and War* (Shubik, 1975).
2. Intermodal Surface Transportation Efficiency Act of 1991.
3. In 1997, N. Mineta was a corporate executive with Lockheed Martin IMS. He is referred to here because he used to be the mayor of San Jose, California, was a member of the US

Congress for 21 years, and is co-author of the ISTEA. In 2001 he was the Secretary of Transport in the Bush administration.

4. Decision theory, optimization theory and game theory constitute a consistent body of knowledge. Many features of PPP can be discussed either as a 'multivariate decision problem' or investigated in the framework of dynamic programming. However, the gaming perspective offers the most natural way to highlight the common ideas used in these theories and to transfer important conclusions valuable for other perspectives.

Appendix A

Large-scale PPP achievements

A brief overview illustrates well how different approaches have been followed and are sought repeatedly in the PPP project delivery business. Attention is focused on the financing techniques applied in large projects. Further details (such as technical data, institutional involvement, etc.) on the listed facilities can be found elsewhere in the literature (Martinand, 1994; Lederer, 1998; Lockwood *et al.*, 2000; Miller, 2000).

Dulles Greenway, Virginia, USA ($350 million): privately funded toll road in the USA connecting Washington, DC, with its international airport. The professional delivery (completion ahead of time, within budget, with high quality) was not sufficient to prevent or avoid financial troubles. The ridership was overestimated and in the mid-1990s the toll revenues were insufficient to make debt service payments.

SR-91 Highway, Orange County, California, USA ($120 million)*: fully automated vehicle identification and tolling applied. Sources of project funds consisted of funded equity (15%), loans from banks (50%) and other institutional tranches (35%). After a low start in 1996 the project arrived at a solid financial performance.*

Tacoma-Narrows Bridge, Washington, USA ($350 million): with an introductory toll of $4 the project was approved by a 53 : 47% margin in a referendum.

Camino Columbia Tollway, Texas, USA ($75 million): debt service schedule is optimized for backloading in order to fully build up the facility's ridership in the initial years.

Southern Connector Extension, Florida, USA ($150 million): a private consortium has put up half of the total project cost in exchange for significant development rights for land along the corridor. The Florida Department of Transport is actually designing, building and operating the toll facility and also retaining the toll revenues (traffic validation projected insufficient ridership for bond sale in the capital market).

Pocahontas Parkway Connector, Virginia, USA ($330 million): the first construction project under Virginia's innovative Public–Private Transportation Act of 1995 (Annex 8) is a nine-mile-long four-lane toll-road. Private bond investors with a state contribution less than 10% have financed it. The investors will be repaid solely by tolls from users of the facility.

Highway 407 Toronto, Ontario, Canada ($2700 million): the largest BOT project in the Western hemisphere until 1997 involved consortium equities, senior and subordinated debts and short-term bank loans to be refinanced with long-term bonds after completion. It is the world's first fully electronic toll highway. The original DBFO financing model was abandoned and the road construction portion of the project was financed directly by the Ontario government (it could finance the same debt at a substantially lower cost than the competing proposals could).

Vasco da Gama Bridge, Portugal ($1000 million): the concession company covers one-third of the costs, the second third comes from the EU as a grant, and the European Investment Bank supports the project with a 20-year project loan.

Cross Israel Highway, Israel ($1100 million): a consortium of local banks, insurance companies, provident and pension funds covers the lion's share of the cost with a contribution funded by overseas institutional investors. The government mitigates the risks and reimburses the operator for the revenue shortfall if the traffic forecast does not meet the accepted forecasts.

Rosario Victoria Bridge, Argentina ($380 million): the toll facility is funded through non-recourse toll-based project finance. The government of Argentina contributes in excess of $200 million towards the total project costs.

Melbourne City Link, Australia ($1200 million): infrastructure equity placements and revenue bonds were issued and combined with significant non-recourse financing on toll revenue basis. The Australian government supported the financial closure with tax-free infrastructure bonds for the investors.

Metro Manila Skyway, Philippines ($1800 million + $800 million): toll-revenue-based BOT financing works successfully.

Delhi-Noida Toll Bridge, India ($96 million): the BOT project is privately financed on a non-recourse basis from projected toll revenues. The sponsor's equity-to-debt ratio is 30 : 70. Roughly 30% of the project debt comes from the IBRD credit line.

ORLYVAL, France (no cost data): private concession for the rail link between Orly airport and the Paris city subway network. The project was a commercial failure due to the overestimation of ridership levels, despite its technical, ecological, operational and legal excellence. The planning experience had been subjected to the discipline of the market. Had the project been the responsibility of a public body the failure would have been covered up.

Eurotunnel, France–UK (undisclosed, < it$15,000 million): the financing scheme, under the terms of a 55-year concession established by the private developers in the 1980s, collapsed well before completion. Difficulties arose from the long construction period and the low level of capital (less than 15%). Consecutive interventions of the French and British governments helped the project to survive but the revenues seem to be inadequate in the long run. The project serves as an example of the absolute necessity of public involvement in large-scale facilities.

M1 Motorway, Hungary ($300 million): the first BOT project in the CEE region financed by private developers and financing institution with the leadership of the European Bank for Reconstruction and Development. Exceptionally high tolls collected at gates were calculated to cover all construction, operation and debt service costs. After completion in 1995 the reduced corridor traffic demand, low domestic solvency, legal challenges and controversial political debates resulted in the financial collapse of the operator. A government buyout and introduction of the national vignette system eliminated private involvement in 2000.

M5 Motorway, Hungary ($400 million): the second private BOT motorway development in Hungary with a substantial public contribution (including the right of tolling on a section constructed earlier by the state) and conservative financing plan based on conservative traffic estimation has operated successfully since 1997. Because of the inconsistency with the national motorway vignette system the government has restrained extension of the route, and buyout was under negotiation in 2001.

Appendix B: economic games described in the literature

Common property resources
(Aliprantis and Chakrabarti, pp. 58–62)

There are n members of a community that have independent access to use a common property resource (e.g. a fishing ground). Let the player i use r_i amount of the resource, the total resource used being:

$$R_i = \Sigma r_i.$$

The cost to player i of getting r_i depends on the amount r_i and on the amount $R - r_i$ used by the other players, as well. This function $C(r_i, R - r_i)$ is assumed to be such that marginal cost increases monotonically with the total amount of the resource used. The player receives from r_i a utility $u(r_i)$. As the amount of r_i consumed increases, the value of an additional unit of r_i falls, while marginal utility at 0 is greater than marginal cost.

The individual player is concerned about the impact of his consumption on the resource only via his cost and ignores the cost imposed on others. In the Nash-equilibrium the pay-off

$$u_i(r_i) - C(r_i, R - r_i)$$

turns out to be equal for all players with a consumption of R^*/n. The total pay-off of the community can be defined now as:

$$n\ [u(R/n)—C(R/n,\ R - R/n)].$$

Depending on the actual structure of the function C the 'social optimum' R^{**} (the maximum of the total pay-off) may often occur at a consumption level $R^{**} < R^*$. Common properties are often exploited beyond the level that is most desirable from a collective viewpoint.

Note: selection of the cost function in a PPP game might be particularly difficult when the use of the analogous public facility has traditionally been accounted inappropriately.

Partnership game
(Aliprantis and Chakrabarti, pp. 110–112)

Two partners P_1 and P_2 (having resources $R_1 < A$ and $R_2 < A$) are interested in financing a project of total cost A ($R_1 + R_2 \geq A$). If they co-operate and succeed, each receives W; otherwise zero. Partner P_1 moves first in period 1 by committing $r_1 (0 < r_1 \leq R_1)$ to the project. P_2 has to decide whether he makes up the difference

$r_2 = A - r_1$ in the second period for $W(0 < r_2 \leq R_2)$. Both know that their opportunity cost of contributing r_i to the project is r < unknown expression > i2 and partner P_1 discounts his pay-off in the second period by the discount factor δ.

The analysis results in the optimal commitments for partner P_1 as $A - W\frac{1}{2}$ and for the partner P_2 (constrained by P_1) as $W\frac{1}{2}$.

Note: private partners looking for a coalition in a PPP project may apply this game. P_1 has an apparent constraint to move first but a certain advantage of enforcing P_2 to come up with the highest reasonable commitment. He is interested in underestimating the value of the project.

Financing a project
(Aliprantis and Chakrabarti, pp. 152–154)

A project is widely known as viable but only a developer (D) knows that it can be worth H or L after an investment of I. Because D cannot finance the project with his own funds he needs to get a venture capitalist (C) to whom he can offer an equity stake $0 \leq e \leq 1$ in return for investing in the project. The investment takes place in the first phase of the game and the return is realized in the second phase. The rate of return i on the project can be taken as equal to or greater than the current rate of interest or the opportunity cost of capital and

$$H - I < L - I > (1 + i)\, I.$$

At the time the offer is made C knows only that the project is worth H with probability p, and L with probability $1 - p$. C will reject the offer e unless

$$p(eH - I) + (1 - p)(eL - I) \geq (1 + i)\, I.$$

Consequently, D's offer e must be slightly higher than $e^* = I\,(2 + i)\{pH + \ (1 - I)L\}^{-1}$. Because e does not convey any information about the probability p, this offer is better than e_H corresponding to the worth H. Since the developer has an incentive to offer the lowest stake to C he suggests that the project is worth H. The capitalist knows this and does not believe any message given by D but looks for independent estimations.

It may happen that the worth L is lower than needed to attract the investment

$$H - I > (1 + i)\, I < L - I$$

and at the estimated level of p

$$p(eH - I) + (1 - p)(eL - I) < (1 + i)I.$$

In this case the game does not start, since C will be better off rejecting any offer. Unfortunately, this occurs even if the project is actually worth H. D can circumvent the breakdown if he knows that actually $p = 1$ and the expected return will be

$$\alpha(H - I) \geq (1 + i)I.$$

Now he can offer to borrow I from C, who will take the offer because he knows that D will make such a proposal only if he knows that the project is worth H.

Note: some players may want to keep the worth of the project uncertain and guess overestimated figures to arrive at lower equity stakes for other players. Timing of the venture capital involvement may be crucial. In some cases the roles have to be revised to reach a deal if the project was strongly promoted.

Allocating the tax burden
(Aliprantis and Chakrabarti, pp. 232–235)

There are n potential users of a promoted public facility Υ having a utility $u_i(\Upsilon) > 0$; $(I = 1 \ldots n)$ if the facility is built, otherwise zero. The utility is interpreted as the maximum amount the player would pay for the access. Each player has a net worth of W_i, and the cost K to build the facility is $\Sigma W_i < K$ (in other words, if the community of players wants to finance the project collecting a tax t_i paid by the individuals then it can to do so). The surplus that the community and the individual player derives from building the facility is

$$\Sigma u_i(\Upsilon) - K \quad \text{and} \quad u_i(\Upsilon) - t_i, \text{ respectively.}$$

Correspondingly, the tax revenue has to cover the cost of the facility; the total surplus of the community is the sum of the individual surpluses.

In order to derive a fair tax system and establish an adequate coalition C of the players to finance the facility, the Shapley value-allocation rule can be applied. This rule renders to every player a marginal worth and makes it possible to analyse quantitatively the costs and benefits of the coalition's enlargement if it were necessary to collect K.

Note: quantitative game-theoretical analysis of the coalition enlargement requested to finance a public facility via local taxes can be combined with several social considerations concerning a successful local option tax referendum described elsewhere (Beale et al., 1996).

Discipline the administration
(Nilsson, 1996)

The constitutional entity in charge of national welfare may decide whether a commercial agent under the control of a regulator or a ministerial administration should supply a public service. Given the costs emerging in any case the decision may depend on:

- the utility function of the administration officers involved in the service (they may have additional objectives other than defined by the constitutional entity);
- the idiosyncratic costs billed by the commercial agent (which can hide a part of its profits or benefits to gain particular advantages).

When selecting the utility functions for the society, administration and agent, further assumptions can be applied in the model, such as:

- the availability of information about the agent's costs for the regulator;
- the higher or lower efficiency of the agent;
- price elasticity of the users.

In the case of the road maintenance and improvement industry, parliament may either charge the Minister of Transport to establish and operate a professional administration or entitle the Minister of Finance to contract with an agent surrendered to a regulator. The game-theoretical analysis shows that if the accountability of the commercial agent is appropriate, while the Minister of Transport is assumed to have strong incentives other than prescribed by parliament, then an organizationally separated private management corresponds better to social preferences than an administrative one. Otherwise, if the administration is instructed precisely about its obligations it is preferred over the regulated agent.

Note: debates between the ministries in several countries about the existence or abolition of the Road Trust Fund, and the apparent appetite of the Ministry of Finance to skim revenues collected in the independent Fund and to use them for general purposes, are well-publicized facts and support the use of this game model for other constellations with different assumptions.

Local public service provision by PPP
(Padon, 1999)

Local governments often compete for enlarging and attracting developments. They can influence the developers' choice of residential location with several provisions. Recent philosophical and ideological trends lead to many public services traditionally provided directly by local governments being supplied instead by various privatized arrangements. An extensive-form game played by the developer, the city and the customer-voter with complete and perfect information about a possible development helps to analyse and compare the public policy of service supply and private partnership attitudes:

- the pay-offs are determined by the variables representing the information required by the players and influencing their decisions:
  - the size of development as a percentage of the given population;
  - the ratio of the housing value and tax base in the development to the city;
  - the value of the public service enhancement for an individual household;
  - the status quo property tax rate;
  - the racial minority share.

Numerical simulations show that there is a prevalence of traditional public strategy (of higher levels of budget-provided services) in those communities that are homogeneous in terms of both race and housing values. A privatized pseudo-public goods development strategy seems to be efficient when an attempt is made to redevelop and expand residential bases previously left by the middle-classes.

Note: local road development cost allocation experience has been discussed with respect to PPP alternatives (Giglio, 1997). Suggestions based on phenomenological considerations can be supplemented and confirmed with quantitative game-theoretical calculation.